

MicroVario

The small powerful TEK-Vario with acceleration sensor



Specifications and operating instructions

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1 Introduction

The MicroVario with a built-in high-resolution pressure sensor gives via telemetry a high quality vario, and with the integrated TEK connection also allows the use of compensation nozzles. This results in a vario function which, independent of the "stick thermals", very accurately indicates the real rise and sink. For an exact height measurement when also operating with TEK nozzle a second Pressure sensor is installed, which provides correct altitude data independent of speed.

The built-in 3-axis acceleration sensor with a special sensor system provides information about the loads in flight. Measuring range up to 16 g.

Via our UniDisplay all measured values of the MicroVario can be viewed live. All settings and alarms can of course also be conveniently programmed via the display.

In accordance with our philosophy to support many systems, the MicroVario also speaks the language of

Jeti Duplex (EX),
Multiplex M-Link
Graupner/SJ HoTT
Robbe/Futaba FASSTest S.BUS2
JR DMSS
FrSky
Spektrum (via the Spektrum-Adapter)
PowerBox P2Bus

The telemetry used in the MicroVario only needs to be specified once in the settings. This is done either through our PC program "SM MicroVarioTool" or the UniDisplay. On delivery HoTT is selected.

2 Description of the MicroVario

- high-resolution vario with TEK connection
- Height measurement with second pressure sensor with automatic zero setting after switching on
- integrated 3 axis acceleration sensor up to +- 16 g
- full telemetry support for Jeti Duplex (EX), Multiplex M-Link, Graupner HoTT, Futaba FASSTest S.BUS2, JR DMSS, FrSky, Spektrum and PowerBox Core
- Connector for receiver signal for remote control of certain functions
- Measurement of the receiver battery voltage
- current status is indicated by LED
- Direct viewing of the measured values live with our UniDisplay
- Parameter settings via PC, UniDisplay and telemetry possible
- Language for telemetry values and operation via UniDisplay switchable German - English
- free firmware updates via PC with our USB interface (order no. 2550) or with already existing existing USB connection cables like from Jeti, Multiplex or Graupner possible
- can be used almost everywhere due to its compact size and low weight

2.1 Technical Data

Vario resolution:	0.01 m/s
Vario noise:	typ. less than 0.02 m/s (vario filter on medium)
Resolution (Altitude):	0.1 m
Acceleration sensor:	3 axes up to 16 g
Power supply:	from receiver supply (from 3.6 V to maximum 8.5 V)
Power consumption:	approx. 24 mA
Connections:	1 x connection for telemetry and power supply ("Link") 1 x servo pulse input from receiver COM port for UniDisplay and firmware update Connection for TEK nozzle
Dimensions:	26 (33 with plug) x 12 x 7 mm
Mass:	1.8 g without cable / 5.4 g with 30 cm standard cable

2.2 Measured Values

The MicroVario can record the following measured values. Depending on the telemetry used, however, only some of this may be available at the transmitter.

Designation	Unit	Content
Vario	m/s	Vario value from barometric vario sensor
Height	m	height above the starting point
Altitude gain	m	Altitude change of the last 10 seconds, is recalculated every second; thus a tendency can be detected during thermal circling
Speed	Km/h	Air speed when using a TEK nozzle
Acceleration X/Y/Z	g	Acceleration in 3 axes each max. +- 16 g
Receiver voltage	V	Voltage at supply input "Link"
Servo pulse width	us	servo pulse at the single "Link" connection; can be used optionally to switch between min/ live / and max values
Air pressure	hPa	air pressure measurement of the barometric sensor

3 Operation of the MicroVario

3.1. Installation

Due to the low weight and compact design, installation is unproblematic. Mounting with Velcro tape on a board is quite sufficient.

Position of the three acceleration sensors.



The Z-axis always points down when the MicroVario front points up. The assignment of X and Y depends on the installation direction.

3.2. Meanings of the LED

The MicroVario has a red status LED.

During operation the following signals are available:

LED flickers in the first 5 seconds after power on

➔ ready for manual setting of the used telemetry, see 3.4 Telemetry selection without PC

LED flashes according to the currently selected telemetry, see 3.4 Telemetry selection without PC

➔ Normal state

3.3. Basic Settings

The settings of the MicroVario can be made with our software "SM MicroVario Tool" on the PC or laptop, with our UniDisplay, or via Jetti and HoTT telemetry.

The following settings are important for the MicroVario to measure correctly:

- **"Telemetrie Auswahl"** defines the telemetry used.
- **"Fixe seriennummer"** so that Jetti and Futaba sensors can be interchanged.
- **"Vario Schwelle"** is the threshold for the vario signal via telemetry separated for rise and fall. Only when the climb / descent is greater than the threshold is a Vario tone generated by telemetry.
- **"Vario Ton"** defines whether the vario for climb / descent or both is active. Here the Vario tone can be turned off completely.
- **"Vario Faktor"** specifies the factor by which the values are multiplied by the Vario for telemetry. Normally it is 1.0, so that the true m / s are displayed on the telemetry. In particular cases, however, by using a factor greater than 1 sound for the telemetry can be made more sensitive if the transmitter itself allows no such setting.
- **"Vario Filter"** allows an adaptation of the response speed of the varios "Slow" corresponds approximately to the previous evaluation, "medium" and "fast" are correspondingly faster. Faster, however, always means a more unsteady signal from the vario.

- **"Höhe sanft Nullen"** keeps the height measurement at 0 m as long as the model is still on the ground. This will result in a drift of the sensor, but also a change of the air pressure is suppressed. The altitude measurement on the ground remains always at 0m. The recognition, if the model is on the ground, is done by the vario. In some cases this standstill detection also activated in the air (extremely smooth flight on a slope), then the automatic zero position of the height can be switched off here.

3.4. Telemetry selection without PC

In addition to the PC software and the UniDisplay, there is a further possibility to select the telemetry without the need to change any tools:

- To do this, the MicroVario is connected to the receiver as described in Chapter 4.1. The three-pole Telemetry cable attaches to the telemetry input of the receiver, the optional Rx cable attached to a free servo output.
- At the transmitter, an encoder (switch or stick) is defined for this servo channel.
- In the first five seconds after switching on the MicroVario, as long as the LED is flickering, this Encoder should be moved back and forth at least 3 times.
- The LED then goes out briefly and then the telemetry selection always flashes three times, corresponding to this list:

☐ Futaba	➔	1 x LED, pause
☐ JR DMSS	➔	2 x LED, pause
☐ HoTT Vario	➔	3 x LED, pause
☐ Jeti Ex	➔	4 x LED, pause
☐ M-Link	➔	5 x LED, pause
☐ FrSky	➔	6 x LED, pause
☐ Spektrum	➔	7 x LED, pause
☐ PowerBox P2Bus	➔	8 x LED, pause
☐ HoTT GAM	➔	9 x LED, pause
- ☐ At the desired selection, the encoder is moved twice more
➔ the LED goes on and the new selection is stored.
- ☐ If you switch the power back out before, no change is made.

3.5. Telemetry alarms

These alarms are output at the transmitter via the attached telemetry. Depending on the system a beep will sound and/or a warning by voice output.

- **"Höhe"** (Altitude)
The alarm is activated as soon as the set height is exceeded. After 20 seconds the alarm automatically deleted and only reactivated when the height is exceeded again. Well suited to be used with the approach a certain height of the towing machine or to monitor a height limit.
- **"Speed min und Max"** (Max and Min Speed)
The alarm is active as long as the set speed is exceeded or undershot.
- **"Rx Spannung"** (Rx voltage)
For monitoring the receiver supply. The alarm is active as long as the voltage is below the set voltage threshold.

3.6. Vario

The MicroVario has a high resolution vario sensor with TEK connection.

TEK = Total Energy Kompensation

TEK Nozzle = Nozzle on the aircraft with hose connection to the vario, usually mounted in front of the vertical tail

The Vario can be operated with or without TEK nozzle. Without nozzle the TEK connection remains simply free and measures the ambient pressure.

3.7. Speed measurement via TEK nozzle

As of firmware v1.03, the MicroVario can also measure speed. The only prerequisite for this is a connected TEK nozzle.

The true air speed is calculated from the pressure at the TEK nozzle and reported via telemetry.

The measurement has a very wide measuring range that goes beyond our model building speeds (>1000 km/h). Nevertheless, the measuring accuracy and stability is very high. The measurement error depends primarily only on the quality of the TEK nozzle. If it is 100% compensated like our Braunschweig-type nozzles, the measuring error is minimal.

TEK Vario and speed measurement with only one nozzle and without further devices!

Speed measurement is only possible with the TEK nozzle connected.

3.8. TEK Connection

TEK put simply means the suppression of “stick thermals” by considering the motion of the model. This is achieved by connecting a nozzle (pitot tube) to the TEK connector of the Vario.

For slower models flying at a constant speed the simple Vario without the TEK nozzle is often sufficient. Here the error due to control inputs is not so great and the thermals can clearly be seen with the Vario. The cleaner and slower you are flying the better the Vario does without the TEK nozzle.

With dynamic models or even with less than perfect flying style the Vario is always indicating, which is not showing the true rise and fall of the model but only a controlled speed or altitude change i.e. the “stick thermals”. We are indeed interested in the thermals, but only whether you are flying in rising, neutral, or falling air mass. If possible regardless of how the aircraft just moved.



Order No. 2780 angled version



Order No. 2781 straight version

Exactly this behaviour is achieved by using a TEK nozzle with the Variometer. This nozzle produces a motion dependent pressure drop simultaneous with increasing speed, a rise which compensates for the height loss. So you press on and the Vario without TEK would indicate sinking. The TEK nozzle compensates for this by indicating the increasing speed available for climbing. The conversion of height (potential energy) into velocity (kinetic energy) is therefore with the TEK nozzle no longer incorrectly displayed as sinking or climbing.

Ideally, with full compensation, it will always show the current sink rate of the aircraft in quiet air. This of course is not a constant value but depends on the speed, buoyancy, and other factors.

The MicroVario has a TEK connection on the front side. The hose of the TEK nozzle is simply plugged on to this. In principle all types of TEK nozzle can be used, including the well-known Nicks nozzle or the Brunswick nozzle. The nozzles differ in sensitivity to sliding and angle of attack and the possible compensation factor. We have on offer a nozzle of the Brunswick type. This is very insensitive to the angle of attack and allows complete compensation. In addition the compensation can easily be adjusted. For details on our nozzle, please visit our homepage.

3.9. Vario output via telemetry

The tone generation of the Vario (except the "old" Jeti Telemetry without EX) is always done in Telemetry transmitter or the Telemetry Box. The vario value transmitted by the MicroVario in m/s is used for this. However, not all transmitters allow the sound output to be adjusted to hide certain areas or adjust the sensitivity of the acoustic output.

For this reason, the MicroVario has the following setting options, which influence the variables for all telemetry:

- **"Vario Schwelle"** indicates the response thresholds for the vario signal via telemetry. Only if the rise / fall is greater than the threshold is the vario value transmitted, and a vario tone generated, by telemetry. Below the threshold the value 0.0 m/s is transmitted.
- **"Vario Ton"** determines whether the vario is active during climb / descent or both. Here the Varioton can also be switched off completely.
- **"Vario Faktor"** determines the factor by which the values from the vario are multiplied for telemetry. Normally it says 1.0, so that the real m/s are shown on the telemetry. However, in some special cases a factor greater than 1 can be used to make the sound output of the telemetry more sensitive if the transmitter itself does not allow such a setting.
But you have to take into account that the variometer value displayed and recorded in the transmitter does not corresponds to reality.
- **"Vario Filter"** enables the response speed of the Vario to be adjusted, "slow" corresponds approximately to the previous evaluation, "medium" and "fast" are correspondingly faster.
Faster, however, always means a more unsteady signal from the vario.

Example:

- “Vario climbing threshold” is set to 0,5 m/s “Vario sinking threshold” is set to -1.0 m/s
- “Vario tone” is set to “on”

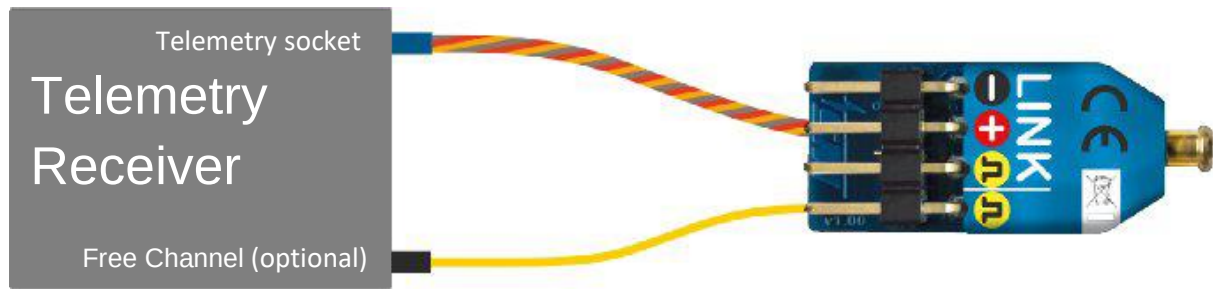
➔ if the model rises faster than 0.5 m/s, the value is sent

➔ if the model rises or sinks more slowly, 0 is sent

If the Vario is always required to be transmitted, set the Vario thresholds to 0.0 m/s and the Vario sound on “up/down”.

4 Connections

4.1. Connection as telemetry sensor



The MicroVario is connected directly to the telemetry port using the three-pin cable with blue plug.

The separate orange cable with only one lead can optionally be connected to a free receiver channel, see 4.2.

For HoTT, M-Link, Robbe/Futaba S.BUS2, JR DMSS, FrSky, Spektrum and PowerBox Core, further sensors are connected directly via Y-cable.

With Jeti Duplex, additional sensors are connected via the Expander E4 or plugged in to separate telemetry inputs of the receiver.

4.2. Using the receiver channel

Optionally, depending on the selected "Rx control" option, the MicroVario can be controlled remotely. For this purpose a second connection between the desired free receiver channel and the single orange line on the telemetry cable of the MicroVario is necessary.

If none of these options are used, the individual receiver connection of the MicroVario can simply remain free.

4.2.1. Switched min and max values per receiver channel

The option "Rx control = Min/Live/Max" is used to switch between the transmission of live, Min and Max values via telemetry. This allows you to read, for example, the maximum altitude from the flight even for telemetries that do not record maximum values in the transmitter itself.

At the transmitter a 3 position switch is programmed for this free channel, which switches the channel between the following values:

- 100 % for the minimum values (switching point < 1.3 ms servo pulse width)
- 0 % for the live values
- + 100% for the maximum values (switching point > 1.7 ms servo pulse width)

During operation, the 3-step switch on the transmitter can be switched back and forth 3 times to reset the minimum and maximum values.

5. Using the UniDisplay

The UniDisplay and MicroVario are connected with the cable supplied with the display. The slot is marked with "COM" on the MicroVario. The display is supplied with power from the MicroVario. The MicroVario must be powered either via a connected receiver or can be supplied with power directly with a receiver battery.



Menu:

After switching on first the menu is activated. The menu options can be selected with the **"Plus"** and **"Minus"** buttons, and the appropriate item selected with **"Enter"**.



Live data display:

- **"Minus"** changes between live / MIN / MAX values.
- **"Esc"** switches back to the menu.

In the upper right corner you can see the time elapsed since power on. "Vario" shows the internal temperature and next to it the air pressure. Below this are the three acceleration values X | Y | Z



Setup:

The menu for all settings of the MicroVario appears here.

In the second line the firmware version of the MicroVario and the serial number are displayed.

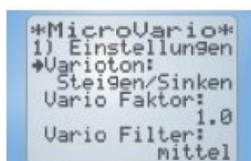
Depending on the telemetry used, there is another menu item 3 for telemetry specific settings.



Settings Screens:

The settings of the MicroVario are summarised here.

Use the **"Plus"** and **"Minus"** keys to select the menu items, and with **"Enter"** the corresponding point can be selected. The arrow then becomes a point and the selected value can be changed with **"Plus"** and **"Minus"**. A press on **"Esc"** or **"Enter"** saves the change. See also 3.3 for the individual points.





MicroVario alarms:

The alarms that can be set here apply to all telemetry versions. Depending on telemetry, alarms can also be set directly in the transmitter. Please see the Notes for the individual telemetry systems.

When the arrow is in the left column and the appropriate menu option activated with "Enter", the value of the alarm can be set.

After the arrow is moved right and the menu option activated with "Enter" the alarm can be activated ("+") or deactivated ("-") with "Plus" or "Minus".



M-Link addresses:

For transfer of MicroVario measured values by M-Link, addresses can be assigned here for the display on Multiplex transmitter. Each address may only be assigned once to any attached M-Link sensor, including the M-Link receiver. The Bus system ceases to function with multiple assignments.

If a value is not to be transmitted chose the address "--". This value comes after the highest permissible Address 15.



Jeti EX values:

Here, all measured values can be transferred to the Jeti transmitter or the JetiBox Profi and can be individually selected. Fewer measured values mean faster transmission, so you can hide the unneeded values here.



Spektrum Init:

The address specifies the assignment in the Spektrum adapter. Here is only important that with Several SM telemetry sensors on the Spektrum adapter no address is assigned twice. Normally the default is OK.

All spectrum sensors supported and operated by the MicroVario can be selected or deselected individually. Each sensor type may only occur once.



Futaba Init:

The compatibility mode for the Robbe Telemetry Box can be activated here.

6 Telemetry operation

The MicroVario supports telemetry via Jeti Duplex (EX), Multiplex M-Link, Graupner HoTT, Robbe/Futaba S.BUS2, JR Propo DMSS, FrSky, Spektrum and PowerBox Core.

The telemetry operation is similar for all supported systems, live data is displayed on the transmitter or on external display, and with Jeti Duplex and HoTT the MicroVario can also be operated from the transmitter. If the system output is a spoken voice, then this is also supported by the MicroVario.

The alarm output depends on the telemetry. In some systems, the MicroVario generates the alarm, in others the thresholds are set directly on the transmitter. Please follow the instructions below.

6.1. Jeti Duplex

The MicroVario is a full-fledged telemetry sensor for Jeti Duplex 2.4 GHz systems. All measured values can be transmitted live to the ground and displayed on the Jeti transmitter. Also, the Jeti Expander E4 for connection of up to 4 sensors is supported.

The connection to the Jeti Duplex receiver is made with the supplied patch cable between the slot "Link" on the MicroVario and "ext." on the Jeti Duplex receiver. With the new REX receivers it must be ensured that the telemetry slot is set to "JETIBOX/Sensor" or, in the case of central boxes, to "Telemetry Input". This setting can be made from the transmitter in the device overview.

10.1.1. Ex Telemetry with the Jeti transmitter and the JetiBox Profi

The MicroVario transmits the data via Jeti EX telemetry to the DC/DS-12/14/16/24 transmitters or to the JetiBox Profi. These data values can be displayed on the transmitter, and voice outputs and alarms can be set. Using the built-in emulation of the JetiBox display, the MicroVario can also be operated and set from the transmitter as described below.

If one MicroVario is replaced by another, the telemetry values must be re-read in the DC/DS-12/14/16/24 or ProfiBox. Each MicroVario has its own serial number and the Jeti EX distinguishes between the various devices on it.

If the MicroVario is to be exchanged, the option "fixed serial number" can be activated. Each MicroVario then sends the serial number 12678 and the devices can be exchanged.

The following measured values are displayed in Jeti EX mode. The values are individually selectable via the PC program or the UniDisplay. The Jeti transmitters can receive a maximum of 32 values simultaneously. When operating with Expander, or to increase the transfer rate, unimportant values can be hidden here.

- | | | | |
|---------------|---------------------|------------------------|----------------------------|
| • VarioTEK | <i>TEK Vario</i> | • Temperatur intern | <i>Internal temp.</i> |
| • Höhe | <i>Height</i> | • Beschleunigung X,Y,Z | <i>Acceleration X,Y,Z</i> |
| • Höhengewinn | <i>Height gain</i> | • Servo Impuls | <i>Pulse width (servo)</i> |
| • RxSpannung | <i>Rx Volts</i> | • Vario | <i>Vario</i> |
| • Luftdruck | <i>Air pressure</i> | • Speed | <i>Air speed</i> |

6.1.2. Operation of the MicroVario with the JetiBox

< - Mx
v

After the start of transmission the JetiBox is changed to Mx for the attached sensors..

Firmware v1.02
SM MicroVario

A press the on the ▼ key changes to the MicroVario initialisation screen then the measured data is displayed.

A 4.93 V Rx
221.8 m +12.8 m/s

As soon as the first screen appears, the different data screens can be selected with presses of ◀ and ▶ keys.

A simultaneous long pressure on the keys ◀ and ▶ changes between the display of Live / MAX / MIN values.

In the top left of the screen there is an indicator of the current active data screen and/or the status of the MicroVario:

:

A first data screen, following screens have B, C, etc

* recording running

✓ maximum values are indicated

◀ minimum values are indicated

< Hoehe Alarm >
(Aus) 200m

A press of key ▼ changes to the settings. Again with the keys ◀ and ▶ the different screens and the desired point selected.

Hoehe Alarm
(Ein) < 200m >

After a further press of key ▼ the selected value can then be changed (keys ◀ and ▶). With a simultaneous pressure on ▲ and ▼ the alarm is switched on/off (Ein/Aus).

Changed settings are only stored with the move back to the selection level with ▲.

6.1.3. Alarms

When operating on the Jeti transmitter modules with the display of data on the simple JetiBox all alarms and Vario sounds are generated directly from the MicroVario. All relevant settings are therefore made on the MicroVario.

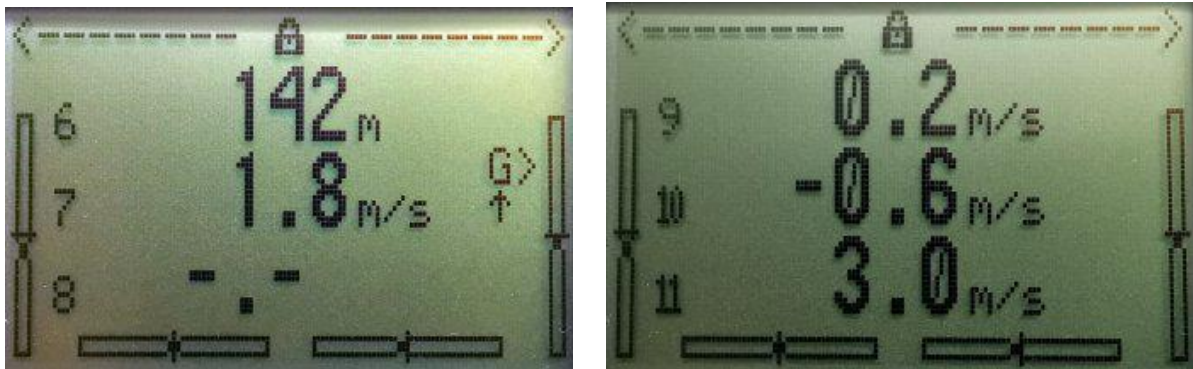
The JetiBox Profi and the Jeti transmitters in Jeti EX mode can actually generate the alerts and sounds. These are then set in the JetiBox or in the transmitter. Alarms that are set in MicroVario are additionally output.

6.2. Multiplex M-Link

The MicroVario is also a fully-fledged telemetry sensor for the Multiplex M-Link 2.4 GHz system. The Measured values from the MicroVario can be transmitted live to the ground and displayed directly on Multiplex ROYAL pro or COCKPIT SX transmitters or the external telemetry screen displays.

The connection to the M-Link receiver is made directly with the supplied connection cable as described in Section 4.1.

On the ground the data is displayed directly on the Multiplex RoyalPro or COCKPIT SX transmitter.



The settings for telemetry are made either with the UniDisplay (see also Chapter 5) or with our "SM MicroVario Tool" software on the PC. The addresses for the display on the Multiplex Remote control (line in which the respective value is displayed), can be freely selected here.

10.2.1. Alarms

With M-Link, all alarms are generated directly by the MicroVario. All settings are therefore made on the MicroVario.

6.3. Graupner HoTT

The MicroVario is also a full-fledged telemetry sensor for the Graupner HoTT 2.4 GHz system. The Measured values can be transmitted live to the ground and shown directly on the display of the HoTT transmitter.

The connection to the HoTT receiver is made with the supplied patch cable between the slot on the MicroVario and "T" on the HoTT receiver.

6.3.1. Alarms

The MicroVario supports both the text mode and the digital mode of the HoTT system.. In both operating modes, all adjustable alarms of the MicroVario are indicated on the transmitter by beeping or voice output.

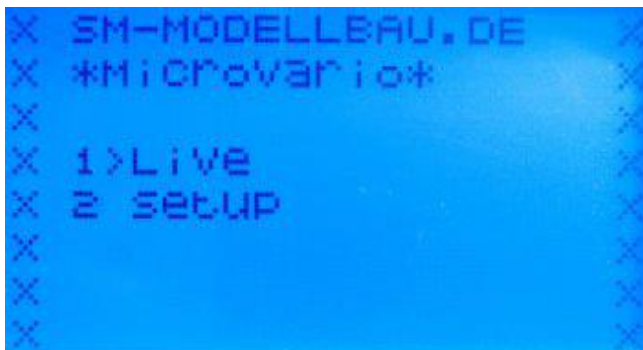
6.3.2. Vario

The HoTT transmitters fade out the range from -0.5 m/s to 0 m/s in the sound output. In order to still have a continuous vario tone, the MicroVario skips this range.

For a more sensitive response of the acoustic output, a "Vario factor" of 2 to 4 should be set in the MicroVario. The result is a good resolution acoustic output without a hole around the zero point. You have to accept, however, that the Vario value displayed and recorded in the transmitter does not correspond to reality.

6.3.3. Text Mode

You can access the text mode via the Telemetry menu and "Settings, display". With the left keypad of the transmitter, the "Vario" can now be called up for the MicroVario using the "up" and "down" keys. With one click right, you switch from the receiver data to the text display of the MicroVario.



Operation is done with the right touchpad of the transmitter. Structure and content are completely identical to the screens of the UniDisplay, see also chapter 5.

Here you can also set all alarms that are indicated on the transmitter by beeping or voice output.

The operation appears somewhat sluggish in text mode, because the data is not updated as often via telemetry.

6.3.4. Digital mode

The MicroVario transmits the data in the "Vario" module format so that the transmitter can display the values in the corresponding screens. From the standard display of the transmitter, the "left" and "right" keys of the left touchpad are used to call up the digital mode. The "up" and "down" keys can now be used to display the Data of the MicroVario and the "Vario" can be called. With the "Left" and "Right" keys of the left touch pad you can now change between each screen.



Some values are assigned differently by the MicroVario:

- **Vario in m/3s:** shows the value "height gain", the height difference in the last 10 seconds

In order to also be able to display the speed when using a TEK nozzle, the MicroVario with HoTT must be configured as a GAM module. Only here does the value "Speed" exist and can also be announced by the transmitter.

6.4. Futaba S.BUS2

The MicroVario can also be used with the Robbe/Futaba FASSTest telemetry as an S.BUS2 sensor. The MicroVario is connected to the S.BUS2 slot of the receiver like any other sensor.

With the S.BUS2 servo data and sensor values can be connected on the same data line. But since but the servo data is much more important than the sensor values, we strongly recommend you strictly separate the servo data from the sensor values. All servos are connected to the S.BUS1 connector of the receiver, all sensors to the S.-BUS2. This means that a sensor can never interfere with the data for the servos, even in the event of an error.

If the MicroVario is nevertheless to be operated together with the servos on S.BUS2, it is essential that a connection cable order no. 9110 is used between MicroVario and S.BUS2! Thus, with this cable the sensor is decoupled from the bus to such an extent that it is impossible to influence the servo data.

6.4.1. Registration on the Transmitter

To use the MicroVario with the S.BUS2, it must first be registered to the transmitter, like all S.BUS2 sensors. To do this, the "Link" connection of the MicroVario is connected to the "S.I/F" socket on the transmitter. Some transmitters like the T14SG do not provide power at this socket, then the MicroVario must be supplied via a Y-cable with a separate battery. The MicroVario behaves like a Robbe/Futaba sensor and is integrated into the system in exactly the same way. Please note also the manual of the transmitter.

However, the MicroVario currently occupies six Robbe/Futaba sensor values to be able to display all measured values. Some measured values cannot be displayed with the correct unit, here you have to remember the assignment for the T14SG. With the T18MZ, you can rename the sensors and thus change the assignment to be more comprehensible.

Using the T1 4SG as an example, the individual steps of the registration process follow here:

When the MicroVario is connected to the transmitter and is supplied with power, the item "Registration" in the sensor menu is called. This registers the sensor in the transmitter and free slots are assigned. Save this sensor and transmitter assignment.

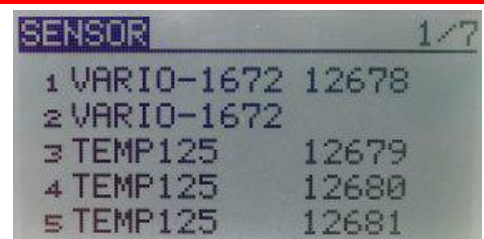
It is absolutely necessary to click on "Registration"!

"Read again" and the other options Mean the MicroVario does not start correctly!



In order to display all values, the menu item "Login" on the transmitter must be activated six times in a row. The registration is finished when the message "Sensor already present" appears.

When the registration of all sensors is finished, the sensor list shows like this:



8 slots are occupied by 7 MicroVario sensors:

Reg	Sensor Name	Slots	ID Fixed serial no.	Values displayed by Microvario		Examples
1	VARIO-1672	2	12678	Hohe Vario	<i>Height Vario</i>	123 m 1.81 m/s
2	TEMP125	1	12679	Beschleunigung X	<i>Acceleration X</i>	100°C =1.00g
3	TEMP125	1	12680	Beschleunigung Y	<i>Acceleration Y</i>	100°C =1.00g
4	TEMP125	1	12681	Beschleunigung Z	<i>Acceleration Z</i>	100°C =1.00g
5	TEMP 125	1	12682	Höhengewinn	<i>Height gain</i>	7°C =7m
6	TEMP 125	1	12683	Empfangsqualität	<i>Reception quality data from the receiver in steps</i>	25/50/75/100
7	SPEED 1677	1	12684	Geschwindigkeit	<i>Speed</i>	123 km/h

Now the MicroVario is connected to the receiver and the telemetry display of the transmitter is called up.

Here again using the T14SG as an example:

TELEM.MONI		1/2
1 HOHE +142 m VARIO-1672	3 TEMPERAT +21°C TEMP125	
1 VARIO +1.80m/s VARIO-1672	7 TEMPERAT +100°C TEMP125	

TELEM.MONI		2/2
4 TEMPERAT -62°C TEMP125	5 TEMPERAT +303°C TEMP125	
6 TEMPERAT +18°C TEMP125		

6.4.2 Alarms

In principle, with the S.BUS2 the alarms are defined in the transmitter. The MicroVario has no way to directly to trigger an alarm at the transmitter.

6.5. JR Propo DMSS

The MicroVario can also be used with the JR Propo DMSS Telemetry. The MicroVario is connected like any other sensor to the sensor slot of the receiver and transfers the following data:

- air pressure, altitude, vario (sensor address 0x03 "Pressure / Altitude")

No more sensors occupying the same addresses can be connected. For the free addresses, further sensors can easily be plugged into the receiver in parallel to the MicroVario with a Y cable.

We tested the connection with the XG8 transmitter firmware version 0001-0017 and the RG831B receiver.

6.5.1. Presentation on the Transmitter



The image shows a monochrome LCD display from a transmitter. At the top, it displays '# 1', '2:41', and '6.40'. Below this, there is a table-like structure with three rows. Each row has a label on the left, a numerical value in the middle, and a status indicator on the right. The first row is 'HÖHE' with '+142.0m' and 'AUS'. The second row is 'VARIO' with '+1.8m/s' and 'AUS'. The third row is 'DRUCK' with '943.9hPa' and 'AUS'. The status indicators are 'AUS' followed by a dashed line.

Label	Value	Status
HÖHE	+142.0m	AUS
VARIO	+1.8m/s	AUS
DRUCK	943.9hPa	AUS

All values can be displayed directly, the order on the display is freely selectable.

6.5.2. Alarms

In principle, the alarms are defined in the transmitter with JR DMSS. The MicroVario has no way to directly trigger an alarm at the transmitter. All alarm thresholds and also the Varioton generation are therefore specified in the transmitter.

6.6. FrSky

With FrSky Telemetry all receivers with "S.Port" connection are supported. For telemetry operation the receiver must be bound in D16 mode!

The telemetry works with FrSky transmitters with FrOS as well as with all transmitters with OpenTx.

The MicroVario is connected like any other sensor to the sensor slot of the receiver and transmits the following data:

- **Höhe** *Height*
- **Vario** *Vario*
- **Beschleunigung** *Acceleration*
- **Höhengewinn** *Height gain*
- **Vario** *Vario*
- **Speed** *Speed*

"Alt"

"VSpd"

"AccX" "AccY" "AccZ"

Adresse 0x0880 in Meter

Adresse 0x08B0

"ASpd", unit can be changed in the transmitter from kts to km/h or m/s

This vario with address 0x08B0 has 0.01 m/s Resolution.

The following settings are required for correct display :

- "Unit" = m/s
- "precision" = 0.00
- "Conversion" = 1.0

As source for the vario then this value should be used because the higher resolution allows a finer tone gradation.

SENSOR4		27 Mei 11:19
1.80m/s		
Name	Vari	
Typ	Sensor	
ID	08B0	22
Quelle	External module	
Einheit	m/s	
Präzision	0.00	
Umrechnung	1.0	
Offset	0.00	
Auto Offset	☐	
Nur Positiv	☐	

The sensor address (ID) of the MicroVario in the FrSky system can be freely selected in the settings. No further sensors occupying the same addresses can be connected. For the free addresses additional sensors are simply plugged into the receiver with a Y-cable parallel to the MicroVario.

6.6.1 Registration on the transmitter

After connecting the MicroVario to the S.Port of receiver, on the transmitter under "Telemetry" select "Start Sensor Search" and the list of sensors is queried.

Afterwards they can be renamed and are available for all OpenTx functions like voice output, alarms, display etc.

TELEMETRIE				27 Mei 14:08
Sensoren		Wert	ID	
1: Alt	*	142.5m	245	
2: VSpd	*	1.8m/s	245	
3: HGaw	*	19m	245	
4: Vari	*	1.80m/s	245	
5: AccX	*	0.21g	245	
6: AccY	*	-0.62g	245	
7: AccZ	*	3.03g	245	
8: TRSS	*	112dB	248	

6.6.2. Display on the Transmitter

All values can be displayed directly, the order on the display is freely selectable.

Here the sensors with a free address have already been marked with a name, e.g. the sensor 0x08B0 as the Vari.



6.6.3. Alarms

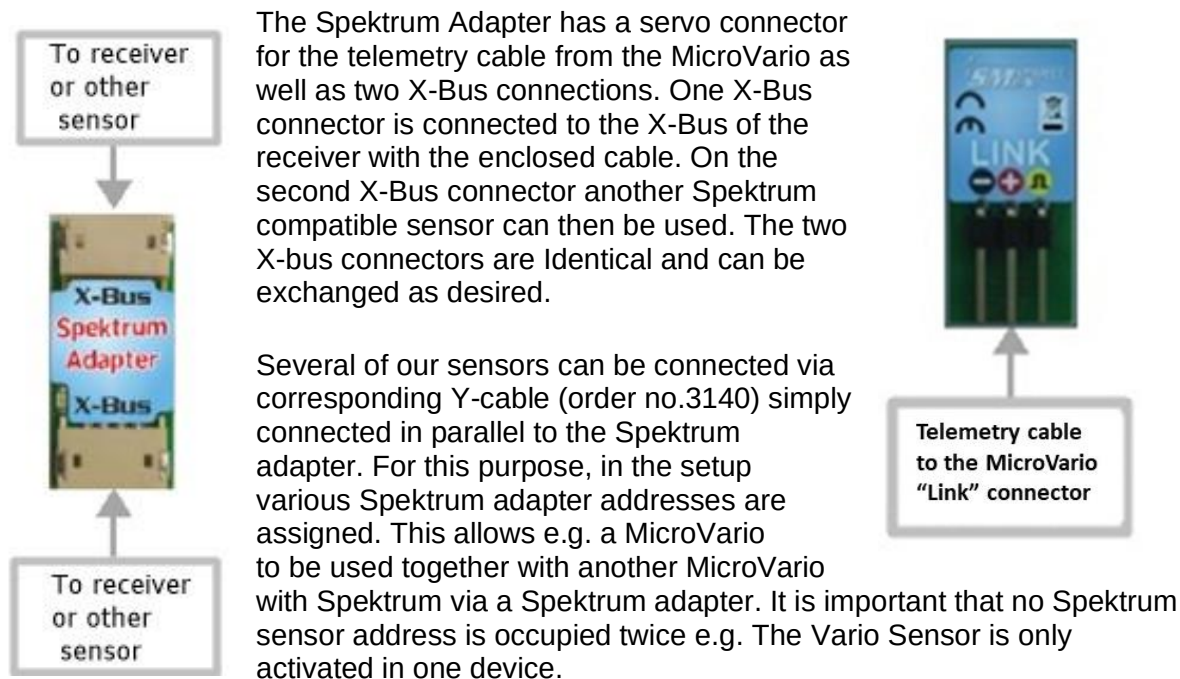
In principle, with OpenTx the alarms are defined in the transmitter. The MicroVario has no way to directly to trigger an alarm at the transmitter. All alarm thresholds and also the Varioton generation are therefore stored in the transmitter by default.

6.7. Spektrum

For spectrum telemetry, the MicroVario is connected to the receiver via our spectrum adapter order no. 9120. This adapter translates the telemetry values into the Spektrum X-Bus format.

6.7.1 Spektrum Adapter

The Spektrum adapter is compatible with all telemetry-capable Spektrum receivers and with the TM1000 System. Currently the receivers with integrated telemetry send the data slightly slower than the TM1000, so there is some longer delay in the vario tones. Therefore, in the MicroVario set the Vario Filter to "fast".



Registration at the Spektrum Transmitter:

The query for the connected sensors with Spektrum comes only once after switching on the receiver. Here the Spektrum adapter responds immediately with the sensors it has learned during the last operation. This means that after every change in the configuration, i.e. after deleting or activating a Spektrum sensor in the MicroVario, the system must be first switched on to learn the new sensor types. The Spektrum Adapter can then only register the changes correctly on the receiver when it is switched on for the second time.

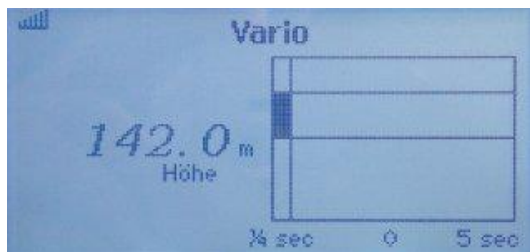
LED of the Spektrum adapter:

During startup the integrated LED turns on and off slowly once. Afterwards it flickers during operation when data is transmitted via XBus.

Each time a sensor is logged on or off, the LED slowly goes on or off to confirm the process.

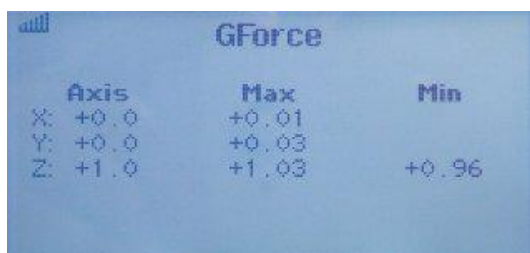
6.7.2. Display on the Transmitter

The MicroVario can display two different Spektrum sensor types. These can be individually deactivated to avoid double assignments on the X-Bus. The data can be processed in the transmitter in exactly the same way as with original Spektrum sensors, i.e. voice output, display, alarms, and storage are possible.



Vario Sensor ("Vario")

- Höhe *Height*
- Vario



Beschleunigungssensor *Acceleration sensor ("GForce")*

- Beschleunigung *Acceleration* X, Y, Z
- Minimum and maximum values



Airspeed

- Höhe *Altitude*
- Geschwindigkeit *Speed* (only with TEK nozzle)

6.7.3. Alarms

In principle, Spektrum defines the alarms in the transmitter. The MicroVario has no way to directly to trigger an alarm at the transmitter. All alarm thresholds and also the Varioton generation are therefore stored in the transmitter by default.

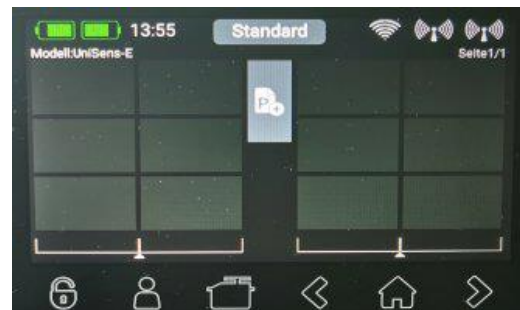
6.8. PowerBox Core P2Bus

The MicroVario can also be used with PowerBox Core Telemetry. Currently the MicroVario supports the full transmission of all telemetry values and the switching between German / English and meter /feet units. A setting menu of the MicroVario is not yet installed.

6.8.1. Display on the Transmitter



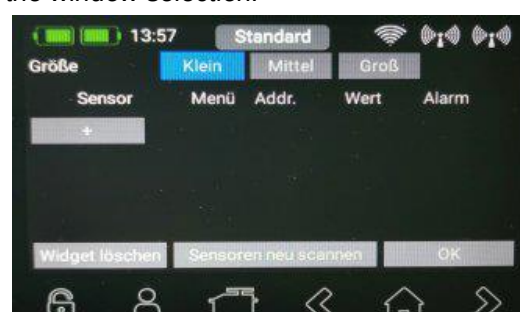
Empty start screen .



A long press on the free area brings up the window selection.



After clicking on the desired window, with "Telemetry" ...



... a widget will be created. The + opens the sensor list.



Now the desired value can be selected



This widget can also have multiple values displayed alternately, just insert +.



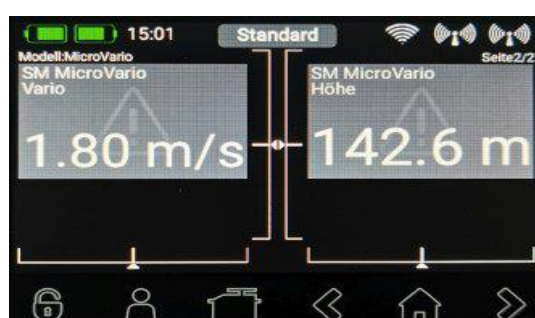
A fully occupied telemetry site .



Alarms are defined directly in this widget in the transmitter



A long click on the sensor in the widget also Shows the address of the sensor. This can also with X → Y be changed to allow several similar sensors to be connected.



The values can also be displayed in large format



English designations and units after Changeover in the system. The sensors switched have to be rescanned in the sensor Widget.

6.8.2. Alarms

In principle, PowerBox defines alarms in the transmitter. The MicroVario has no way to directly to trigger an alarm at the transmitter. All alarm thresholds and also the Varioton generation are therefore defined in the transmitter.

7 The "SM MicroVario Tool" software

On our website you will find the free software "SM MicroVario Tool" in the Software & Updates section with the following functions:

- reading and changing settings of the MicroVario via (almost) any USB interface
- live data display with a simulated UniDisplay over our USB interface (Order No. 2550)
- automatic online search for new firmware for the MicroVario from our website
- installing firmware updates on the MicroVario

If you hover your mouse over the buttons, you will see short help texts for operation.

7.1. Connecting the MicroVario a PC

Editing the settings and updating firmware on the MicroVario is possible with almost any USB interface cable. Our own USB interface (No. 2550) with 4 wire connection is suitable as well as many update cables of current transmitters / receivers / controllers. These usually have a 3 core cable and a standard JR servo plug / socket.

Currently the following cables have been tested:

- | | | |
|---|---|---|
| SM-Modelbau-USB Interface
part no 2550 | ➔ | > is directly connected to the MicroVario COM port
> No further adapters needed.
> First plug in the cables then click "connect" in the software
> With older SM USB interface cables it may happen that the MicroVario does not react at all, so the LED does not light up either. In this case the MicroVario must be separately powered. First click on "connect" and connect a battery to the link input of the MicroVario during the search. Jeti |
| Duplex "USBa" USB adapter | ➔ | > connect to the LINK port of the MicroVario.
> No further adapters needed.
> First click "connect" in the software then connect the Jeti cable to the MicroVario |
| Multiplex "USB PC cable"
85149 | ➔ | > connected to the LINK port of MicroVario.
> V cable with Rx battery for power required.
> First connect MicroVario with V cable to the MPX cable, then click "connect" in the software and finally plug the battery into the V cable |
| Graupner "USB interface"
part no. 7168.6 | ➔ | > connected to the LINK port of MicroVario.
> adapter cable 7168.S required.
> First click "connect" in the software then connect the 7168.S adaptor cable to the MicroVario |

Third-party cables with servo plugs are connected directly to the MicroVario at "Link". When the USB cable has a servo socket use the normal cable from the MicroVario as adapter.

With all cables that connect to the MicroVario via "link", you must click on "Verbinden" (Connect) in the "SM MicroVario Tool" prior to making the connection and activating the power supply. Otherwise the MicroVario starts up normally and can no longer be addressed by the PC

7.2. Settings

The software opens with the "Settings" tab. Here the settings of the MicroVario are read and displayed. When a value is changed, the data must be written to the device with the button "geänderte Einstellungen übertragen" (transfer changed settings).

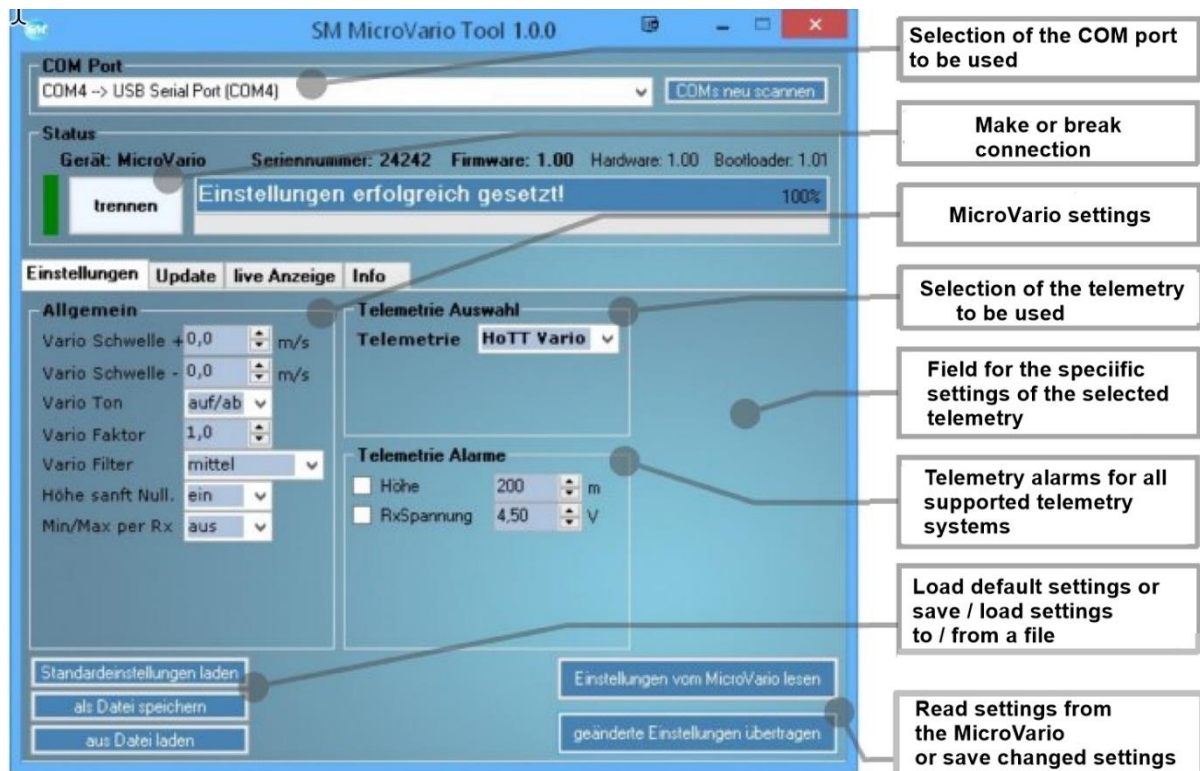


Illustration 1: Settings tab

7.3. Firmware update for the MicroVario

A new firmware is simply uploaded to the MicroVario via (almost) any USB interface. A corresponding file with the update can be downloaded free of charge from our "SM MicroVario Tool" and transferred to the MicroVario.

When the PC software "SM MicroVario Tool" is started, the program automatically searches for newer firmware on our server. If a newer file is found, it can be set to be automatically loaded onto the MicroVario. This ensures that the MicroVario always remains up to date.

Procedure

- The MicroVario must be connected to the "SM MicroVario Tool".
- On the "Update" tab search for a new firmware.
- With the **“ausgewählte Firmware Datei übertragen“** button, the update will start.
- Subsequently the MicroVario reports back with the new version number.

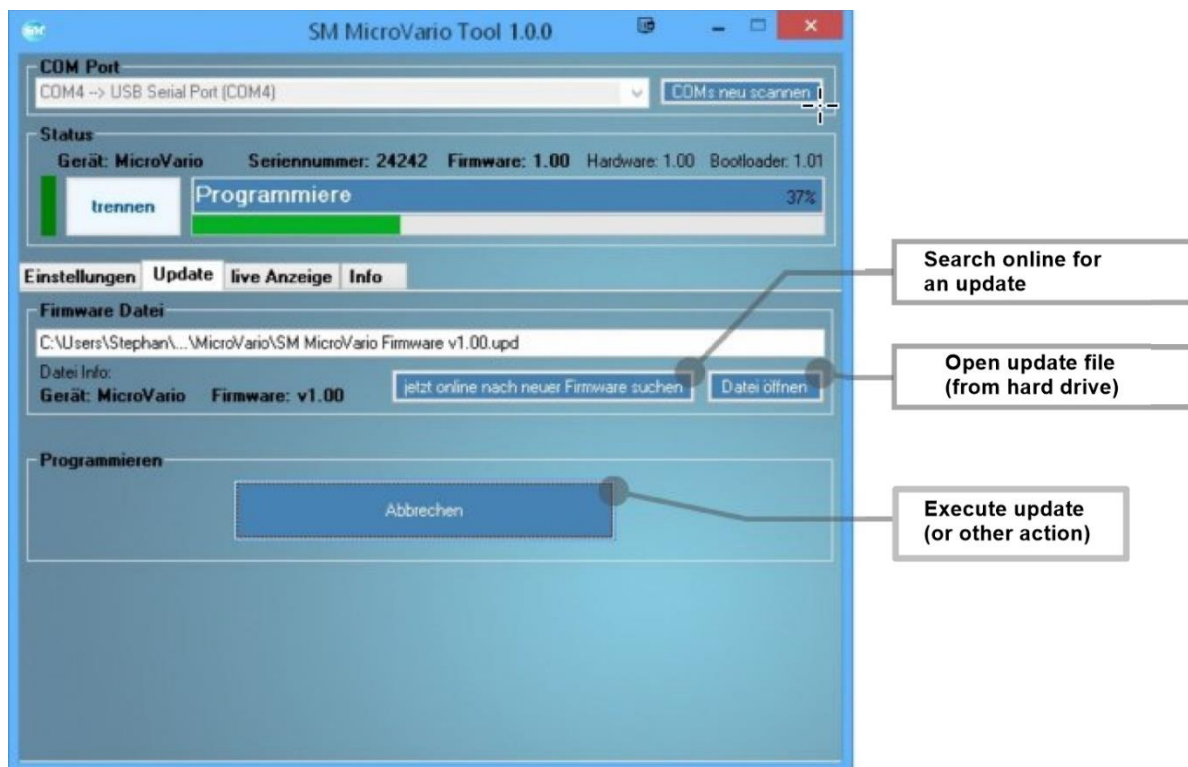


Illustration 2: Update tab

7.4. Live Access to the MicroVario

If the MicroVario is connected to the PC with our USB interface, the MicroVario can be accessed directly with "live display" of our software. The display is identical with the live operation of the UniDisplay. See also chapter 5.

The USB interface is identical to the one used so far with the UniSens-E or the GPS-Logger interface. It can be ordered individually under the order no. 2550.

The "live display" requires a separate power supply for the MicroVario on the "Link" connection. (E.g. Directly with a 4-cell receiver battery.)
This feature is only possible with our USB interface 2550. Other USB cables are not suitable.

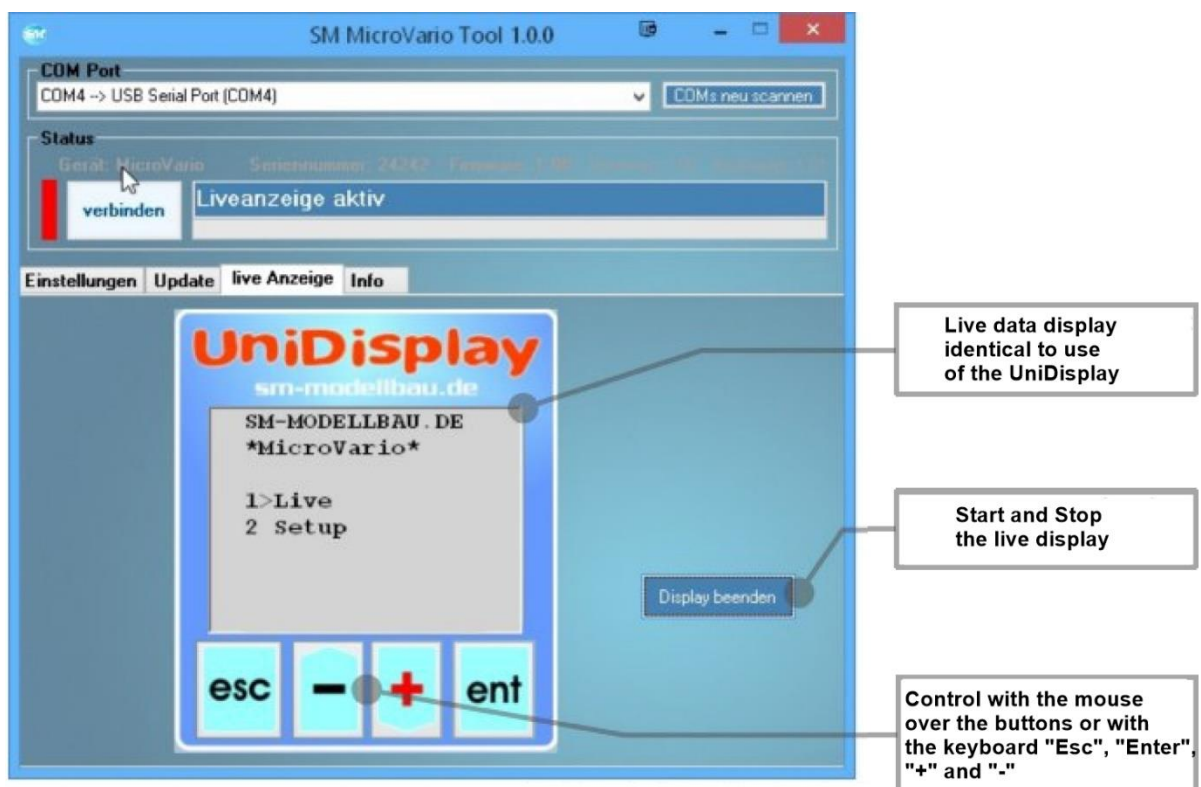


Illustration 3: Live data tab

8 Version History

Here you will find all firmware versions and the changes to the previous version.
The version of your MicroVario Firmware can be read with our software "SM MicroVario Tool" or the UniDisplay.

Version	Date	Comments
1.00	06.2020	First version – sales start
1.01	06.2020	1. soft zeroing was always on 2. soft zeroing was missing in the UniDisplay settings
1.02	07.2020	1. jeti: there are now two varios: VarioTEK and Vario (without TEK nozzle both are equal) 2. vario filter made faster (medium corresponds to the previous fast)
1.03	10.2020	1. speed measurement via TEK nozzle installed
1.04	03.2021	1. the MicroVario can now be switched from German to English. 2. Futaba: Speed is now registered as SBS-01TAS because not all transmitters know the Robbe 1677 sensor 3. HoTT: Text mode could not be reached in the GAM setting.

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